

David Finkelstein, ca. 1958. [Photo by Herbert S. Sonnenfeld; courtesy David Finkelstein.]

The geometry of spacetime outside an imploding star is that of Schwarzschild, and thus the star's implosion could be described using Finkelstein's new reference frame. Now, Finkelstein's new frame was quite different from the reference frames we have met previously (Chapters 1 and 2). Most of those frames (imaginary laboratories) were small, and all portions of each frame (top, bottom, sides, middle) were at rest with respect to each other. By contrast, Finkelstein's reference frame was large enough to cover simultaneously the regions of space-time far from the imploding star, the regions near it, and all regions in between. More important, the various parts of Finkelstein's frame were in motion with respect to each other: The parts far from the star were static, that is, not imploding, while the parts near the star were falling inward along with the imploding star's surface. Correspondingly, Finkelstein's frame could be used to describe the